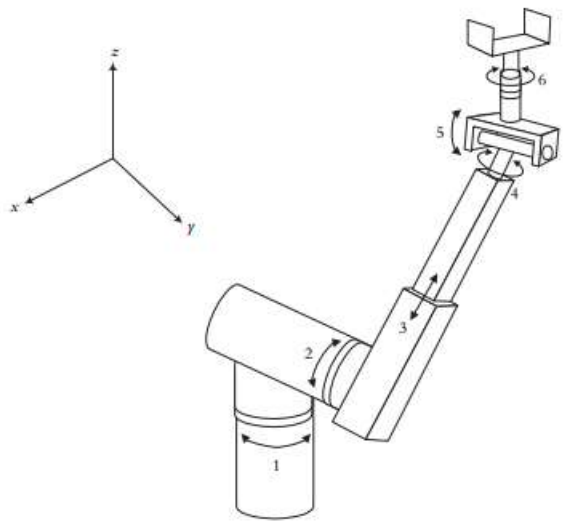
EBE5204 ASSIGNMENT 1 (Individual) 2026

1. For the following 6 degrees of freedom Stanford Arm medical robot



   a) Assign the coordinate frames based on the D-H representation. [5]
   b) Construct the D-H parameters table. [5]

2. With the aid of a well-labeled diagram, explain how a Proportional-Integral-Derivative (PID) controller can be implemented to regulate the force applied by a surgical robot during a surgical procedure. Your response should highlight the control system architecture, the role of feedback, and how each component of the PID controller contributes to precise force regulation. [10]

Due Date: 05/03/2026